

Biological interpretation model for prioritized IPF candidates

Each module links robust bulk signals, machine-learning priority, single-cell context, and a cautious oligonucleotide development angle.

Macrophage/myeloid activation

Genes/axis: SPP1
Cell context: GSE135893 macrophages; GSE136831 macrophage/myeloid
Oligonucleotide angle: candidate knockdown or pathway-modulation target after functional validation

Stromal matrix remodeling

Genes/axis: COL1A1;COL1A2;COL3A1;POSTN;COL14A1;ASPN
Cell context: myofibroblast/stromal and mesenchymal compartments
Oligonucleotide angle: upregulated mRNAs are conceptually compatible with siRNA/ASO knockdown screening

Stromal antioxidant/metabolic loss

Genes/axis: GPX3
Cell context: fibroblast and myofibroblast populations
Oligonucleotide angle: not a direct knockdown candidate; suggests restoration or pathway-protection hypothesis

Epithelial/ciliary remodeling

Genes/axis: TPPP3;DNAI1;TEKT1;RSPH4A;MNS1;RPGRIP1L
Cell context: epithelial and ciliated/axonemal programs
Oligonucleotide angle: candidate epithelial program for mechanistic validation, not yet a direct therapeutic target

Higher-confidence miRNA-target axis

Genes/axis: hsa-miR-375 -> CLDN1/MNS1/RPGRIP1L
Cell context: bulk miRNA/mRNA inverse evidence; cell context requires targeted validation
Oligonucleotide angle: miRNA replacement or target-axis perturbation hypothesis requiring isoform-specific validation